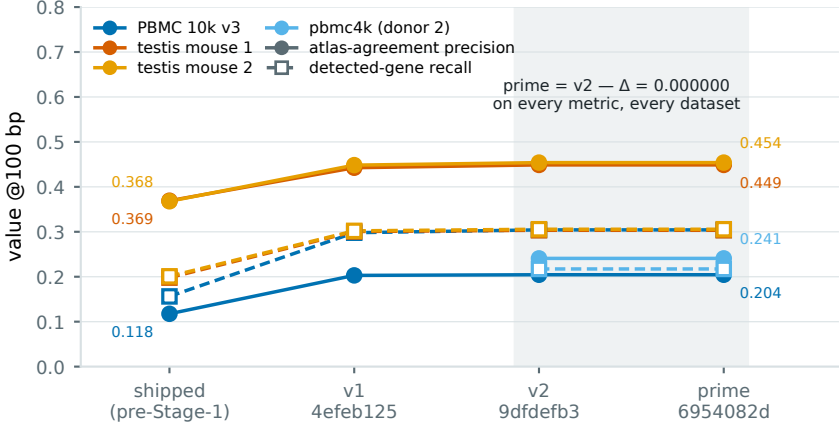


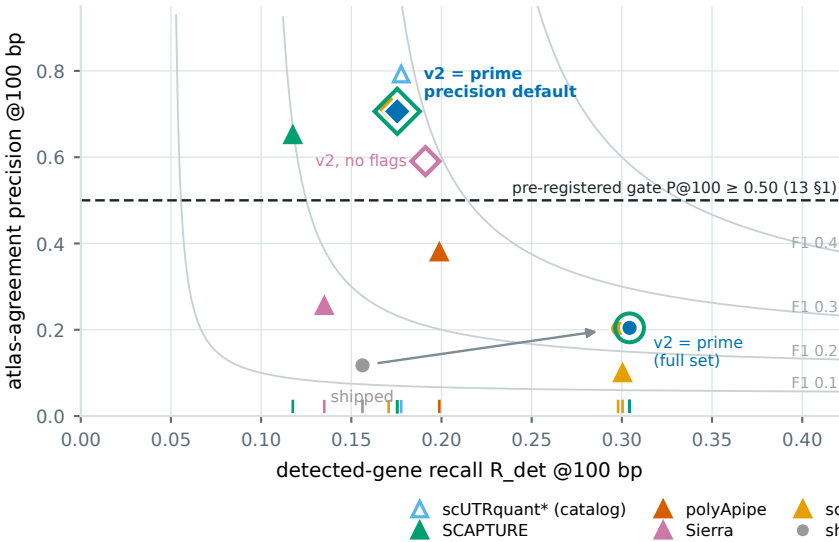
a Progression on the full call set

the arm every version emits. shipped's code has no internal-priming veto; v1 and v2 were given one with --ip-filter; prime turns it on by itself. Solid = precision, dashed = recall.



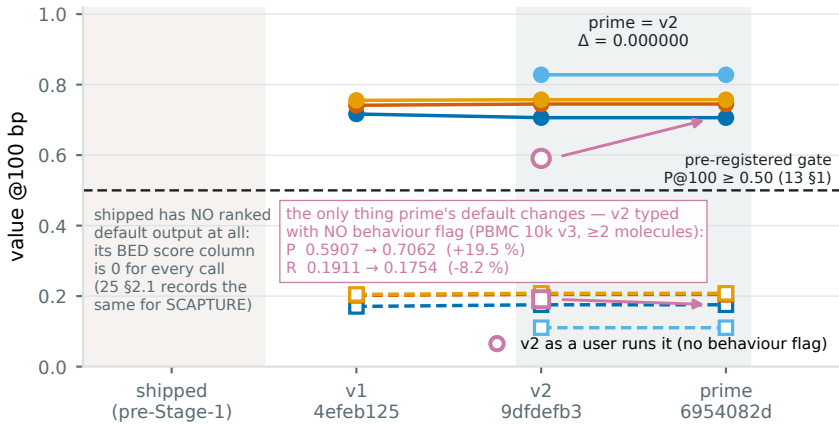
c Against the field — PBMC 10k v3

circles = full call set, diamonds = the precision default; grey arrows run shipped → v1 → v2 → prime, and prime is a RING because it lands exactly on v2. The hollow pink diamond is v2 typed with no behaviour flag — panel b draws the move from it to prime, which is too short to arrow at this scale. Ticks along the bottom = the 3-seed genic-shuffle null (mean) of each call set. scTail is absent: not runnable on this BAM (R1 = 28 bp).



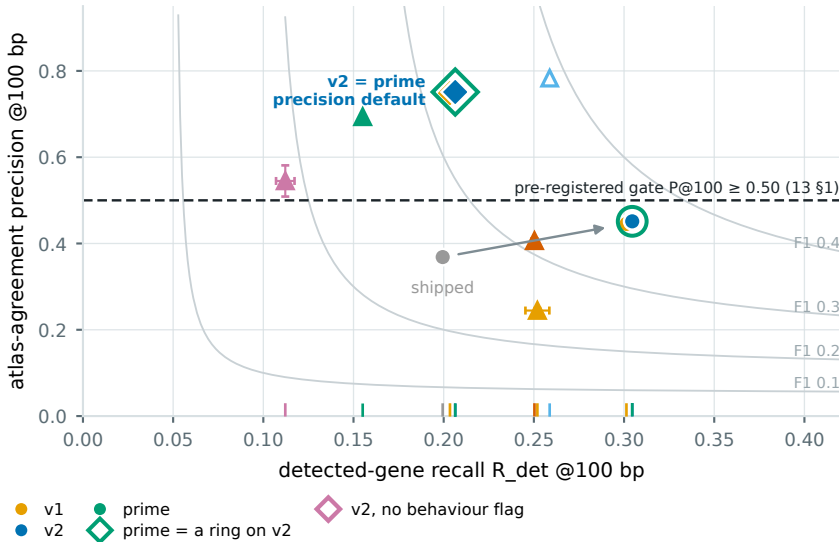
b Progression at the pre-registered default

tier-1 n IP-pass n ≥ 2 clip molecules (12 CORRECTION, 13 \$1). Hollow markers = v2 run with no behaviour flag, i.e. what a user gets by typing nothing. Δ precision v1 → v2: PBMC -0.0105 · m1 +0.0036 · m2 +0.0018 — small, and drawn small.



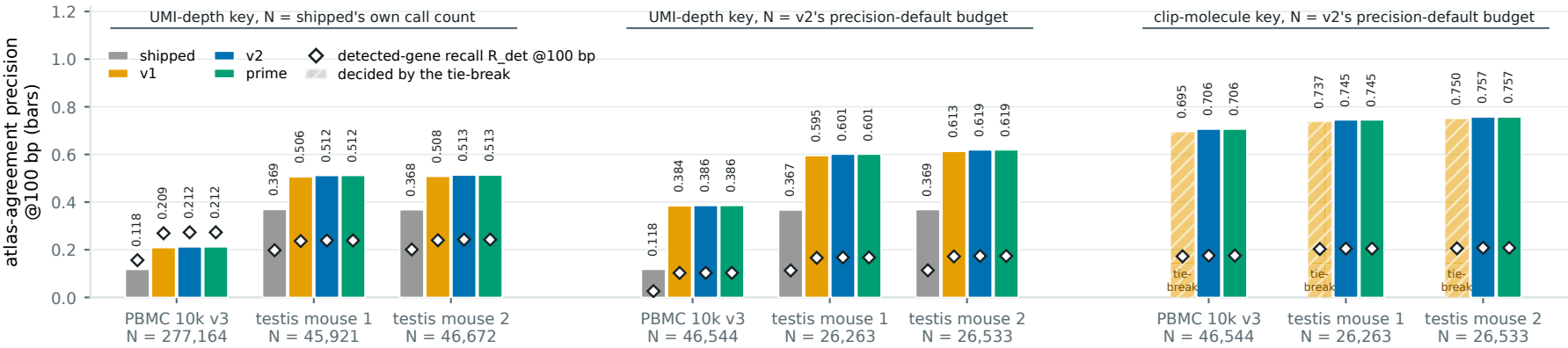
d Against the field — GSE104556 testis

two mice, mean $\pm$ range. Every mouse arm of every version ran with the IP veto, so there is no 'no flags' point here (19 \$3, 24 \$2). Ticks = the 3-seed genic-shuffle null (mean) of each call set. SCAPTURE ran on mouse 1 only.



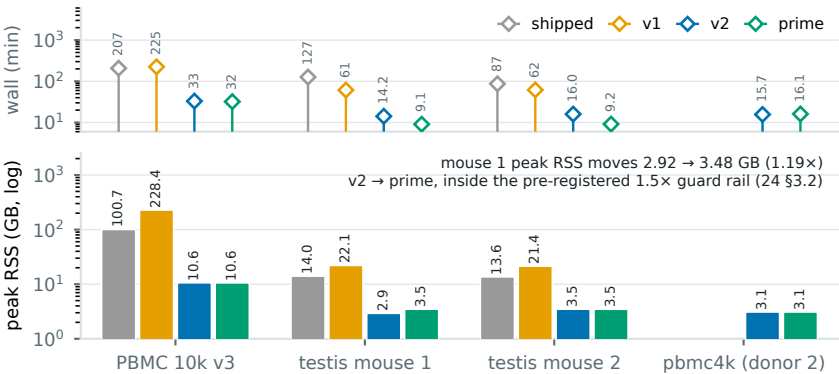
e Matched call count — each version's own ranking, truncated to a common N

the honest way to compare call sets of different sizes (25 \$1). Two keys, because the four versions do not share one: shipped has no clip-molecule column at all. Hatched = the cut falls inside a tie group and the row is decided by the tie-break, not by evidence (on PBMC v1 keeps 2,131 of 117,243 tied sites) — not like-for-like. Same library, same budget (N = 46,544), same code, only the key differs: clip molecules 0.7062 vs UMI depth 0.3857 — the Stage-1 clip evidence is what makes the ranking work.



f What each step cost — compute

the #97 work is the story: PBMC peak RSS 228.4 → 10.6 GB = 21.5× (280.1 → 11.9 = 23.5× on the flagless pair); prime adds none back. WALL TIME IS A RUN RECORD, NOT A BENCHMARK: the v2 arms ran concurrently, prime's one at a time with BLAS pinned to 1 thread. Both strips are log; wall time is the upper strip.



g Does the atlas agree with the long reads?

PBMC 10k v3, ≥ 2 -molecule output. Bars 1-3 higher is better; bar 4 (Kinnex internal-priming decoy) lower is better. v1 → v2 moves both truths slightly DOWN together — a correctness fix, not a score gain. v2-no-flags → prime moves all four the right way at once, the only change here that does — and still does at matched call count, which these bars are NOT at (n 44,394 / 46,524 / 46,524 / 62,110; cut to prime's N, v2-no-flags reads 0.664 / 0.678 / 0.523 / 0.281).

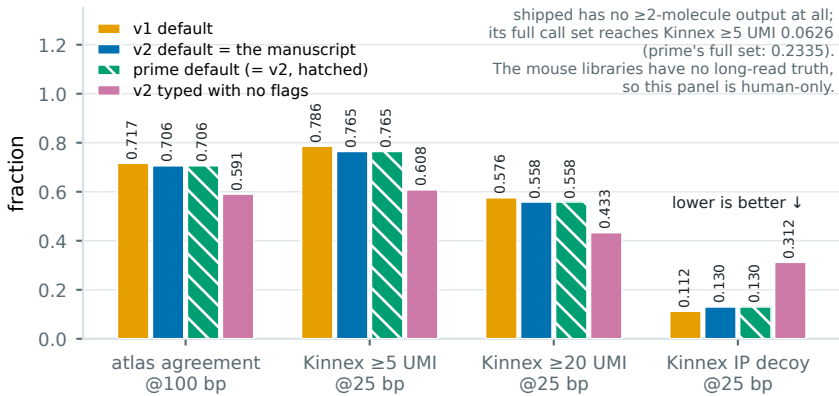


Fig | Four versions of the PeakATail caller on four libraries, scored once. shipped (pre-Stage-1, coverage-only) → v1 (4efeb125, clip-seeded; 15) → v2 (9dfdefb3, + minus-strand IP fix #96 + performance #97; 19, verifier FIXED, pbmc4k added by 26) → prime (6954082d, branch peakAtail-prime). v2 is the manuscript's current record. prime is a development branch: not merged, not the default, and every prime value here is EXPLORATORY with respect to the manuscript's gates (24 \$3.4). Flags (PBMC arm): shipped had no clip evidence, no tiers and no internal-priming veto; v1 and v2 ran --peak-strategy clip_seeded --ip-filter --genome-fasta FA --ip-filter-mode filter; prime ran --peak-strategy clip_seeded --genome-fasta FA — two flags fewer, none added, because --ip-filter-default auto and --ip-filter-mode auto → filter now supply them. HEADLINE: prime's default reproduces v2's manuscript arm byte for byte (md5-identical pas.bed, pas_tier1.bed and precision default on all four libraries), so the pre-registered PRIMARY criterion of 24 \$3.1 ($\Delta P \geq -0.005$ AND $\Delta R \geq +0.010$ AND $\Delta F1 > 0$, on all three gate datasets at once) FAILS with every Δ exactly 0.000000 — no new accuracy on these libraries, and no regression either. What prime's default does change is what a user gets by typing nothing: PBMC $P@100$ 0.5907 → 0.7062 (+19.5 %) for R_det 0.1911 → 0.1754 (-8.2 %), Kinnex ≥ 5 -UMI concordance 0.6081 → 0.7647 and internal-priming decoy rate 0.3122 → 0.1300; prime's set is a strict SUBSET of v2's flagless set (0 added, 68,855 removed). Precision default $P@100$ v1 0.717/0.741/0.755 → v2 = prime 0.706/0.745/0.757 (gate $P \geq 0.50$ PASS on all three); v1 → v2 is a BUG FIX, not an accuracy gain, and PBMC precision goes slightly DOWN with it. PROVENANCE: shipped / v1 / v2 are the runs recorded and verified in 15, 19 and 26, re-scored here with the same scorer (240 recomputed-vs-archived comparisons, 0 mismatches); prime's four arms are NEW runs from an unmerged branch, checked by this benchmark only and not yet through an independent manuscript verification pass. Atlas precision is agreement with PolyASite 2.0, not ground truth; Kinnex truth is a different donor and the mice have none; shipped and v1 were never run on pbmc4k and no such run was fabricated; wall time is not comparable across version sets (f). Every plotted value: results/figures/manuscript/version_progression*.tsv.